

3	Answer: B
	<u>From FBD of block A:</u> $F_{\text{net}} = ma$ $12 + 2.0 \times 9.81 \times \sin 35^\circ - T = 2.0a$ $23.25 - T = 2.0a \quad \text{---} > \text{Eqn(1)}$ <u>From FBD of block B:</u> $F_{\text{net}} = ma$ $T - 20 - 5.0 \times 9.81 \times \sin 20^\circ - 20 = 5.0a$ $T - 36.78 = 5.0a \quad \text{---} > \text{Eqn(2)}$ From Eqn(1), $58.125 - 2.5T = 5.0a \rightarrow \text{Eqn(3)}$ (3) – (2): $94.905 - 3.5T = 0$ $T = 27 \text{ N}$

4	Answer: B
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